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Stat 707 HW2

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In a given day, y out of 10 patients test positive for COVID. Assume tests are independent and each patient has same probability θ for COVID infection.

$y = \text{positive tests}$
 $\text{pmf} = \binom{10}{y} \theta^y (1-\theta)^{10-y}$
 binomial

$\theta = \text{prob of infection}$
 $\text{pmf} = \binom{r+y-1}{y} \theta^y (1-\theta)^r$ where $y = \text{failed (positive test)}$
 negative binomial

a) There are only 10 total patients tested per day (e.g. $y/10$ is not a simplified/reduced fraction)

1. Tests are not repeated until a certain # positive tests are made, n is fixed (binomial)

i. $\hat{\theta}_{ML} = y/n = 0/10 = 0$

ii. $\hat{\theta}_{ML} = y/n = 3/10 = 0.3$

$E(Y/n) = \frac{n\theta}{n} = \theta$ if $Y \sim \text{Binomial}(n, \theta)$

2. Tests are given until y positive tests (failures) - (negative binomial)

i. $\hat{\theta}_{ML} = \frac{r-1}{y+r-1} = \frac{9}{9+0} = 1$

ii. $\hat{\theta}_{ML} = \frac{r-1}{y+r-1} = \frac{6}{6+3} = 0.6$

b) Mean Squared Error: $V(\hat{\theta}) + [B(\hat{\theta})]^2 = E[\hat{\theta}^2 - E(\hat{\theta})]^2 = (E[\hat{\theta} - \theta])^2$

Both estimates are unbiased, $B(\hat{\theta}) = 0$, so $MSE = Var(\hat{\theta})$

1. $Var(\hat{\theta}_{bin}) = Var(Y/10) = \frac{1}{100} Var(Y) = \frac{n\theta(1-\theta)}{100} \rightarrow \frac{\theta(1-\theta)}{10}$

2. $Var(\hat{\theta}_{NB}) = Var\left(\frac{r-1}{y+r-1}\right) = Var\left(\frac{r-1}{y+r-1}\right) = \frac{Var(r-1)}{(r-1)^2} = \frac{Var(y)}{(r-1)^2} = \frac{r(1-\theta)}{(r-1)^2} = \frac{(10-y)(1-\theta)}{81} = \frac{10-10\theta-y+y\theta}{81}$

c) Find MLE for each experiment

1. $\text{pmf} = \binom{10}{y} \theta^y (1-\theta)^{10-y} \rightarrow L(\theta) = \prod_{i=1}^n \binom{10}{y_i} \theta^{y_i} (1-\theta)^{10-y_i}$

$\ell(\theta) = \sum_{i=1}^n y_i \log(\theta) + (10 - \sum_{i=1}^n y_i) \log(1-\theta) + \sum_{i=1}^n \log\left(\binom{10}{y_i}\right)$

$\ell'(\theta) = \frac{\sum y_i}{\theta} - \frac{10 - \sum y_i}{1-\theta} \rightarrow \text{set to 0: } \frac{\sum y_i}{\theta} = \frac{10 - \sum y_i}{1-\theta} \rightarrow \frac{\sum y_i}{\theta} = \frac{10}{1-\theta} \rightarrow \hat{\theta} = \frac{\sum y_i}{10}$

i. $y = 0, \hat{\theta}_{ML} = 0$ SE: $\sqrt{\frac{\theta(1-\theta)}{n}} = 0$

ii. $y = 3, \hat{\theta}_{ML} = 0.3$ SE: $\sqrt{\frac{0.3(0.7)}{10}} = \sqrt{0.021} \approx 0.1449 = 0.14582$

2. $\text{pmf} = \binom{r+y-1}{y} \theta^y (1-\theta)^r \rightarrow L(\theta) = \prod_{i=1}^n \binom{r+y_i-1}{y_i} \theta^{y_i} (1-\theta)^r$

$\ell(\theta) = r \log \theta + y \log(1-\theta) \rightarrow \ell'(\theta) = \frac{r}{\theta} - \frac{y}{1-\theta}$

If we wanted to keep θ constant, we use $1-\theta$, essentially same results but w/ a, b, s swapped in observations

i. $y = 0, r = 10 \rightarrow \hat{\theta}_{NB} = \frac{10}{0+10} = 1$ SE: $\sqrt{\frac{r(1-\theta)}{n}} = 0$

ii. $y = 3, r = 7 \rightarrow \hat{\theta}_{NB} = \frac{7}{10} = 0.7$ SE: $\sqrt{\frac{7(0.3)}{10}} = \sqrt{0.21} = 0.45826$

Bin(3, 0.7)

$\hat{\theta}_{NB} = \frac{r}{r+y} = \frac{7}{10}$

d) 1. Binomial: $\binom{10}{y} \theta^y (1-\theta)^{10-y}$, $\text{Beta}(\theta, 1) \rightarrow \pi(\theta) = \frac{1}{\theta} = 1$

Uniform

Posterior: $\frac{\pi(\theta) P(y|\theta)}{\int_0^1 \pi(\theta) P(y|\theta) d\theta}$

$\rightarrow \frac{\binom{10}{y} \theta^y (1-\theta)^{10-y}}{\int_0^1 \binom{10}{y} \theta^y (1-\theta)^{10-y} d\theta}$

$\rightarrow \text{Beta}(1, 1)$
 $= \frac{\theta^{1-1} (1-\theta)^{1-1}}{B(1, 1)} = 1$

$\rightarrow \alpha \theta^{\alpha-1} (1-\theta)^{\beta-1}$
 $\alpha = y+1, \beta = 11-y$

$\rightarrow \text{Beta}(y+1, 11-y)$

★ i. $y=0$: Mean = $\frac{1}{12}$ SD = 0.07665
ii. $y=3$: Mean = $\frac{4}{12}$ SD = 0.13074

Mean: $\frac{\alpha}{\alpha+\beta} = \frac{y+1}{y+1+11-y} = \frac{y+1}{12}$

SD: $\left(\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} \right)^{1/2} = \left(\frac{11(10-y)y}{18^2} \right)^{1/2}$

Jeffrey's prior or $\sqrt{I(\theta)}$

Likelihood: $\prod_{i=1}^n \binom{10}{y_i} \theta^{y_i} (1-\theta)^{10-y_i} \rightarrow \theta^{\sum y_i} (1-\theta)^{10n - \sum y_i}$

$\ell(\theta) = y \log \theta + (10-y) \log (1-\theta)$

$\ell'(\theta) = \frac{y}{\theta} - \frac{10-y}{1-\theta}$

$\ell''(\theta) = -\frac{y}{\theta^2} - \frac{10-y}{(1-\theta)^2}$

$I(\theta) = -E[\ell''(\theta)] = E\left[\frac{y}{\theta^2} + \frac{10-y}{(1-\theta)^2} \right] = \frac{1}{\theta^2} E(y) + \frac{1}{(1-\theta)^2} E(10-y)$

$= \frac{n\theta}{\theta^2} + \frac{10-n\theta}{(1-\theta)^2} = \frac{10}{\theta} + \frac{10(1-\theta)}{(1-\theta)^2}$

$\rightarrow \frac{10}{\theta} + \frac{10(1-\theta)}{(1-\theta)^2} = \frac{10}{\theta} + \frac{10}{1-\theta}$

$\frac{10(1-\theta)}{\theta(1-\theta)} + \frac{10\theta}{\theta(1-\theta)} = \frac{10}{\theta(1-\theta)}$

Jeffrey's: $\pi(\theta) \propto \sqrt{I(\theta)} = \theta^{-1/2} (1-\theta)^{-1/2}$
 $\propto \text{Beta}(\frac{1}{2}, \frac{1}{2})$

$\alpha = \frac{1}{2}$
 $\beta = \frac{1}{2}$

Posterior: $\propto \theta^y (1-\theta)^{10-y} \theta^{-1/2} (1-\theta)^{-1/2}$

$\propto \theta^{y-1/2} (1-\theta)^{10-y-1/2} \rightarrow \text{Beta}(y+\frac{1}{2}, \frac{21}{2}-y)$

i. $y=0$ Mean = $\frac{0+\frac{1}{2}}{0+\frac{1}{2}+\frac{21}{2}-0} = \frac{1/2}{11} = \frac{1}{22}$, SD = $\left(\frac{(y+\frac{1}{2})(\frac{21}{2}-y)}{(\frac{21}{2})^2} \right)^{1/2} = 0.06013$

ii. $y=3$ Mean = $\frac{3+\frac{1}{2}}{3+\frac{1}{2}+\frac{21}{2}-3} = \frac{7/2}{11} = \frac{7}{22}$, SD = $\left(\frac{(3+\frac{1}{2})(\frac{21}{2}-3)}{(\frac{21}{2})^2} \right)^{1/2} = 0.13445$

Beta(3.5, 7.5)

$$\text{Beta: } \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

$$\text{Mean: } \frac{\alpha}{\alpha+\beta}$$

$$\text{Var: } \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

d) 2. Negative Binomial

Uniform prior $\theta \sim U(0,1)$ $P(\theta) = \frac{1}{1-0} = 1 \rightarrow \text{Beta}(1,1)$

Posterior $\propto \theta^r (1-\theta)^y$ $\rightarrow \text{Beta}(r+1, y+1)$

i. $y=0, r=10$

$\rightarrow \text{Beta}(11, 1)$

$$\text{Mean: } \frac{11}{12}, \text{ SD} = \sqrt{\frac{11}{12^2(13)}} = 0.0766$$

ii. $y=3, r=7$

$\rightarrow \text{Beta}(8, 4)$

$$\text{Mean: } \frac{8}{12} = \frac{2}{3}, \text{ SD} = \sqrt{\frac{32}{144 \cdot 13}} = 0.1307$$

Jeffrey's Prior $\propto \sqrt{I(\theta)}$

$$l(\theta) = y \ln(\theta) + r \ln(1-\theta)$$

$$l'(\theta) = \frac{y}{\theta} - \frac{r}{1-\theta}$$

$$l''(\theta) = -\frac{y}{\theta^2} - \frac{r}{(1-\theta)^2}$$

$$I(\theta) = \frac{y(1-\theta)^{-1}}{(1-\theta)^2} = \frac{y}{(1-\theta)^3}$$

$$-E[l''(\theta)] = E\left[\frac{y}{\theta^2}\right] + E\left[\frac{r}{(1-\theta)^2}\right] = \frac{y}{\theta^2} + \frac{r(1-\theta)^{-1}}{(1-\theta)^2}$$

$$= \frac{y}{\theta^2} + \frac{r(1-\theta)^{-1}}{(1-\theta)^2} = \frac{y}{\theta^2} + \frac{r}{(1-\theta)^3}$$

$$\propto \theta^{-1} (1-\theta)^{-1/2}$$

$\rightarrow \text{Beta}(1, 1/2)$

Posterior $\propto \theta^r (1-\theta)^y \cdot \theta^{-1} (1-\theta)^{-1/2}$

$\rightarrow \theta^{r-1} (1-\theta)^{y-1/2} \rightarrow \text{Beta}(r, y+1/2)$

$$\frac{r(1-\theta)^{-1}}{\theta^2(1-\theta)^3} + \frac{r^2}{(1-\theta)^4} = \frac{r(1-\theta)^{-1}}{\theta^2(1-\theta)^3} + \frac{r^2}{(1-\theta)^4} = \frac{r}{\theta^2(1-\theta)^4} = \frac{r}{\theta^2(1-\theta)^4} = I(\theta)$$

i. $y=0, r=10$

$\rightarrow \text{Beta}(10, 1/2)$

$$\text{Mean: } \frac{10}{10.5} = 0.9523, \text{ SD} = \sqrt{\frac{5}{(10.5)^2(11.5)}} = 0.062798$$

ii. $y=3, r=7$

$\rightarrow \text{Beta}(7, 7/2)$

$$\text{Mean: } \frac{7}{10.5} = 0.66, \text{ SD} = 0.139$$

$$MSE = VCov + [B(w)]^2 = E(\hat{\theta} - E(\hat{\theta}))^2 + E(E(\hat{\theta}) - \theta)^2$$

e) 1. Binomial

a) Uniform prior $\rightarrow \text{Beta}(y+1, 11-y)$ posterior

b) Jeffreys' prior $\rightarrow \text{Beta}(y+\frac{1}{2}, \frac{11}{2}-y)$ posterior

Mean Beta: $\frac{\alpha}{\alpha+\beta}$

a) $\frac{y+1}{12}$ $\xrightarrow{\text{binomial}} B(y) \equiv n\theta = 10\theta \quad \text{Var}(y) = \theta(1-\theta) \cdot 10$

$$MSE\left[\frac{y+1}{12}\right] = \text{Var}\left[\frac{y+1}{12}\right] + \left[E\left(\frac{y+1}{12}\right) - \theta\right]^2$$

$$= \frac{1}{144} \text{Var}(y+1) + \left[\frac{1+2\theta}{12}\right]^2$$

$$= \frac{1}{144} (10\theta(1-\theta) + 1) + \left[\frac{1+2\theta}{12}\right]^2$$

$$= \frac{10\theta - 10\theta^2}{144} + \frac{1 + 4\theta + 4\theta^2}{144} = \frac{1 + 6\theta - 6\theta^2}{144}$$

b) $\frac{y+1/2}{11}$ $MSE\left[\frac{y+1/2}{11}\right] = \text{Var}\left[\frac{y+1/2}{11}\right] + \left[E\left(\frac{y+1/2}{11}\right) - \theta\right]^2$

$$= \frac{1}{121} \text{Var}(y+1/2) + \left[\frac{10\theta + 1/2 - 11\theta}{11}\right]^2$$

$$= \frac{10\theta(1-\theta)}{121} + \frac{1/4 - \theta + \theta^2}{121}$$

$$= \frac{1/4 + 9\theta - 9\theta^2}{121}$$

a) Negative Binomial

2. Uniform prior $\rightarrow \text{Beta}(r+1, 1+y)$ posterior

b) Jeffreys' prior $\rightarrow \text{Beta}(r, y+1/2)$ posterior

$y \sim NB, E(y) = \frac{r(1-p)}{p} \quad \text{Var}(y) = \frac{r(1-p)}{p^2}$

Mean Beta: $\frac{\alpha}{\alpha+\beta}$

a) $MSE\left[\frac{r+1}{r+y+2}\right] = MSE\left[\frac{10-y+1}{10-y+2+2}\right] = MSE\left[\frac{9-y}{12}\right]$

this is surely wrong

$$\text{Var}\left[\frac{9-y}{12}\right] + \left[E\left(\frac{9-y}{12}\right) - \theta\right]^2$$

$$= \frac{1}{144} \text{Var}(y) + \left[\frac{9-r(1-p)}{12} - \theta\right]^2$$

$$\rightarrow \frac{r-rp}{144p^2} + \left[\frac{9-r(1-p)}{12} - \theta\right]^2$$

$$\frac{r-rp}{144p^2} + \left[\frac{9-r+r(1-p)+12\theta}{12}\right]^2$$

b) $MSE\left[\frac{r}{r+y+1/2}\right] = MSE\left[\frac{10-y}{10.5}\right]$

$H_0: \theta = \theta_0$ vs $H_1: \theta < \theta_0$ $\theta_0 = 0.2, 0.4, 0.6$ when $y=0, y=3$

f) p-values (observe & more extreme, under null)

	Null	$H_0: \theta = 0.2$	$H_0: \theta = 0.4$	$H_0: \theta = 0.6$
Binomial	$y=0$	$\sum_{y=0}^0 \binom{10}{y} (0.2)^y (0.8)^{10-y} = 0.10737$	$\sum_{y=0}^0 \binom{10}{y} (0.4)^y (0.6)^{10-y} = 0.00604$	$\sum_{y=0}^0 \binom{10}{y} (0.6)^y (0.4)^{10-y} = 0.0001$
	$y=3$	$\sum_{y=0}^3 \binom{10}{y} (0.2)^y (0.8)^{10-y} = 0.87913$	$\sum_{y=0}^3 \binom{10}{y} (0.4)^y (0.6)^{10-y} = 0.38228$	$\sum_{y=0}^3 \binom{10}{y} (0.6)^y (0.4)^{10-y} = 0.05476$
Negative Binomial	$y=0$	$\binom{9}{0} (0.8)^{10} (0.2) = 0.10737$	$\binom{9}{0} (0.6)^{10} (0.4) = 0.00604$	$\binom{9}{0} (0.4)^{10} (0.6) = 0.0001$
	$y=3$	$\sum_{y=0}^3 \binom{9}{y} (0.2)^y (0.8)^{10-y} = 0.41436$	$\sum_{y=0}^3 \binom{9}{y} (0.4)^y (0.6)^{10-y} = 0.45261$	$\sum_{y=0}^3 \binom{9}{y} (0.6)^y (0.4)^{10-y} = 0.09435$

★ Negative Binomial

we want to

use 1-0 here, $y=0$ results are the same, $y=3$ little different

g) Posterior Probability Using problem 2

	Null	$H_0: \theta = 0.2$	$H_0: \theta = 0.4$	$H_0: \theta = 0.6$
Bin-Unit	$y=0$	0.9141007	0.9963728	0.9999
Bin-Jett	$y=3$	0.1611342	0.7037157	0.9707185
Bin-Jett	$y=0$	0.4676126	0.4487971	0.9999857
Bin-Jett	$y=3$	0.2054427	0.7349255	0.4721323
NB-Unit	$y=0$	0.0000002	0.000	
NB-Unit	$y=3$			
NB-Jett	$y=0$	see binomial		
NB-Jett	$y=3$			

★ these are probabilities on the alternate hypothesis

So for Bin-Unit, $y=0$, $H_0: \theta = 0.2$ vs $H_1: \theta < 0.2$

Suggests there is a 91% chance that θ is less than 0.2, which is similar intuition under all values of θ were equally likely

707hw2 calculations

d

```
set.seed(123)
# Binomial
# Uniform Prior -> Beta(y+1,11-y)
mean(rbeta(10000,1,11)) # y = 0
```

```
## [1] 0.08295777
```

```
mean(rbeta(10000,4,8)) # y = 3
```

```
## [1] 0.3308796
```

```
# Jeffreys Prior -> Beta(y+1/2,21/2-y)
mean(rbeta(10000,0.5,10.5)) # y = 0
```

```
## [1] 0.0454361
```

```
mean(rbeta(10000,3.5,7.5)) # y = 3
```

```
## [1] 0.3194187
```

d

```
# Negative Binomial
# y = 0, so r = 10
nb_uni_prior1 <- rbeta(10000,11,1) # Beta(r+1,1+y)
NBPost1 = data.frame(PostMean = mean(nb_uni_prior1),PostSD=sd(nb_uni_prior1))
nb_jeff_prior1 <- rbeta(10000,10,1/2) # Beta(r,y+1/2)
NBPost2 = data.frame(PostMean = mean(nb_jeff_prior1),PostSD=sd(nb_jeff_prior1))
rbind(NBPost1,NBPost2)
```

```
##      PostMean      PostSD
## 1 0.9173491 0.07603299
## 2 0.9530127 0.06353553
```

```
# y = 3, so r = 7
nb_uni_prior2 <- rbeta(10000,8,4) # Beta(r+1,1+y)
NBPost3 = data.frame(PostMean = mean(nb_uni_prior2),PostSD=sd(nb_uni_prior2))
nb_jeff_prior2 <- rbeta(10000,7,7/2) # Beta(r,y+1/2)
NBPost4 = data.frame(PostMean = mean(nb_jeff_prior2),PostSD=sd(nb_jeff_prior2))
rbind(NBPost3,NBPost4)
```

```
##      PostMean      PostSD
## 1 0.6689716 0.1299602
## 2 0.6644149 0.1398888
```

f

```
y=c(0,3)
theta0 = c(0.2,0.4,0.6)
# BINOMIAL
pbinom(0,10,theta0,lower.tail=TRUE)
```

```
## [1] 0.1073741824 0.0060466176 0.0001048576
```

```
pbinom(3,10,theta0,lower.tail=TRUE)
```

```
## [1] 0.87912612 0.38228060 0.05476188
```

```
# NEGATIVE BINOMIAL
# equivalent for y=0
pbinom(3,9,theta0,lower.tail=TRUE)
```

```
## [1] 0.91435827 0.48260966 0.09935258
```

g

```
## BINOMIAL
# Uniform prior -> Beta(1+y,11-y)
pbeta(theta0,shape1=1,shape2=11,lower.tail=TRUE) # y=0
```

```
## [1] 0.9141007 0.9963720 0.9999581
```

```
pbeta(theta0,shape1=4,shape2=8,lower.tail=TRUE) # y=3
```

```
## [1] 0.1611392 0.7037157 0.9707185
```

```
# Jeffrey's prior -> Beta(1/2+y, 21/2-y)
pbeta(theta0, shape1=0.5, shape2=10.5, lower.tail=TRUE) # y=0
```

```
## [1] 0.9676126 0.9987971 0.9999857
```

```
pbeta(theta0, shape1=3.5, shape2=7.5, lower.tail=TRUE) # y=3
```

```
## [1] 0.2054427 0.7349255 0.9726323
```

```
## NEGATIVE BINOMIAL
# Uniform prior -> Beta(r+1, 1+y)
pbeta(1-theta0, shape1=11, shape2=1, lower.tail=TRUE) # y=0, r=10
```

```
## [1] 8.589935e-02 3.627971e-03 4.194304e-05
```

```
pbeta(1-theta0, shape1=8, shape2=4, lower.tail=TRUE) # y=3, r=7
```

```
## [1] 0.83886080 0.29628426 0.02928148
```

```
# Jeffrey's prior -> Beta(r, y+1/2)
pbeta(1-theta0, shape1=10, shape2=0.5, lower.tail=TRUE) # y=0, r=10
```

```
## [1] 0.0369048429 0.0015866364 0.0000231827
```

```
pbeta(1-theta0, shape1=7, shape2=3.5, lower.tail=TRUE) # y=3, r=7
```

```
## [1] 0.81953258 0.30555239 0.03809081
```

```
# ** Used 1-theta so that results can be compared to binomial results, that is
# theta still indicates the probability of a positive test.
```